ELECTRONIC CONTROL MODULES

TABLE OF CONTENTS

page

ELECTRONIC CONTROL MODULES
STANDARD PROCEDURE - PCM/SKIM PROGRAMMING 1
COMMUNICATION
DESCRIPTION 2
OPERATION 3
CONTROLLER ANTILOCK BRAKE
DESCRIPTION 4
OPERATION 4
REMOVAL 4
INSTALLATION 4
DATA LINK CONNECTOR
DESCRIPTION - DATA LINK CONNECTOR 4
OPERATION - DATA LINK CONNECTOR 4
POWERTRAIN CONTROL MODULE
DESCRIPTION
DESCRIPTION 5
DESCRIPTION - MODES OF OPERATION 5
DESCRIPTION - 5 VOLT SUPPLIES 8
DESCRIPTION - IGNITION CIRCUIT SENSE . . 8
DESCRIPTION - POWER GROUNDS 8
DESCRIPTION - SENSOR RETURN 8
OPERATION
OPERATION 8
OPERATION - 5 VOLT SUPPLIES 9
OPERATION - IGNITION CIRCUIT SENSE 9
REMOVAL 9
INSTALLATION 10
SENTRY KEY IMMOBILIZER MODULE
DESCRIPTION 10
OPERATION 10
REMOVAL 12
INSTALLATION 13
TRANSMISSION CONTROL MODULE
DESCRIPTION 14
OPERATION 14
STANDARD PROCEDURE - TCM QUICK LEARN 17

ELECTRONIC CONTROL MODULES

STANDARD PROCEDURE - PCM/SKIM PROGRAMMING

CAUTION: ASSURE THE DRBIII® IS PROGRAMMED WITH THE LATEST VERSION OF CURRENT SOFTWARE.

NOTE: Before replacing the PCM for a failed driver, control circuit, or ground circuit, be sure to check the related component/circuit integrity for failures not detected due to a double fault in the circuit. Most PCM driver/control circuit failures are caused by internal component failures (i.e. relays and solenoids) and shorted circuits (i.e. pull-ups, drivers, and switched circuits). These failures are difficult to detect when a double fault has occurred and only one DTC has been set.

When a PCM (JTEC) and the SKIM are replaced at the same time, perform the following steps in order:

(1) Program the new PCM (JTEC).
(2) Program the new SKIM.
(3) Replace all ignition keys and program them to the new SKIM.

PROGRAMMING THE PCM (JTEC)

The SKIS Secret Key is an ID code that is unique to each SKIM. This code is programmed and stored in the SKIM, the PCM, and the ignition key transponder chip(s). When replacing the PCM, it is necessary to program the secret key into the new PCM using the DRBIII® scan tool. Perform the following steps to program the secret key into the PCM.

(1) Turn the ignition switch to the On position (transmission in Park/Neutral).
(2) Use the DRBIII® and select THEFT ALARM, SKIM, then MISCELLANEOUS.
(3) Select PCM REPLACED (GAS ENGINE).
(4) Enter secured access mode by entering the vehicle four-digit PIN.
(5) Select ENTER to update PCM VIN.

NOTE: If three attempts are made to enter secured access mode using an incorrect PIN, secured access mode will be locked out for one hour. To exit this lockout mode, turn the ignition switch to the ON position for one hour, then enter the correct PIN. (Ensure all accessories are turned off. Also monitor the battery state and connect a battery charger if necessary).